

Some Interrupt Examples

Hardware Interrupts

Generated by hardware devices to signal the CPU that attention is needed.

Examples:

- Keyboard press/release – Each keystroke triggers an interrupt so the OS can read the keycode.
- Mouse movement/click – The mouse controller sends interrupts when it updates position or button state.
- Timer interrupt – Generated by a hardware timer at fixed intervals (e.g., system clock tick for task scheduling).
- Disk I/O completion – The disk controller signals when a read/write operation finishes.
- Network packet arrival – A network interface card (NIC) raises an interrupt when a packet is received.
- Power button press – Triggers an interrupt to begin shutdown or sleep sequence.
- Printer out of paper – Hardware sensor signals an interrupt to alert the OS.

Software Interrupts (Traps)

Initiated by software, either intentionally (system calls) or due to program errors.

Examples:

- System calls – Programs request kernel services (e.g., reading a file, creating a process).
- int 0x80 on Linux x86 (historical) for syscalls.
- Divide-by-zero error – CPU triggers an interrupt when division by zero occurs.
- Breakpoint interrupt – Debuggers use this (e.g., int 3 on x86) to pause program execution.
- Invalid opcode exception – CPU encounters an instruction it doesn't recognize.
- Overflow/underflow – Arithmetic operation exceeds data type limits.
- Page fault – Access to a virtual memory page not in physical RAM (handled by OS to load page).

Exceptions (Faults, Traps, Aborts)

- Special type of software interrupt caused by the CPU detecting an error condition during execution.
- Examples:
 - Page fault (fault) – Recoverable; OS loads page from disk and restarts instruction.
 - General protection fault – Occurs when a program violates memory protection rules (e.g., accessing kernel memory).
 - Stack overflow – When stack pointer exceeds stack bounds.
 - Alignment check – Accessing misaligned data on architectures requiring alignment.
 - Machine check exception – Hardware failure detected by CPU (e.g., memory parity error).

Maskable vs. Non-Maskable Interrupts (NMI)

Interrupts can also be categorized in to Maskable and Non-Maskable Interrupts

- Maskable Interrupts (can be ignored if CPU's interrupt flag is disabled):
- Most hardware interrupts (keyboard, timer, disk).
- Can be temporarily disabled during critical sections.
- Non-Maskable Interrupts (cannot be ignored):
- Hardware failure (e.g., memory parity error, system overheating).
- Watchdog timer – Critical system monitor alert (common in embedded/real-time systems).
- Power fail – Triggered when power is about to be lost so the system can save state.